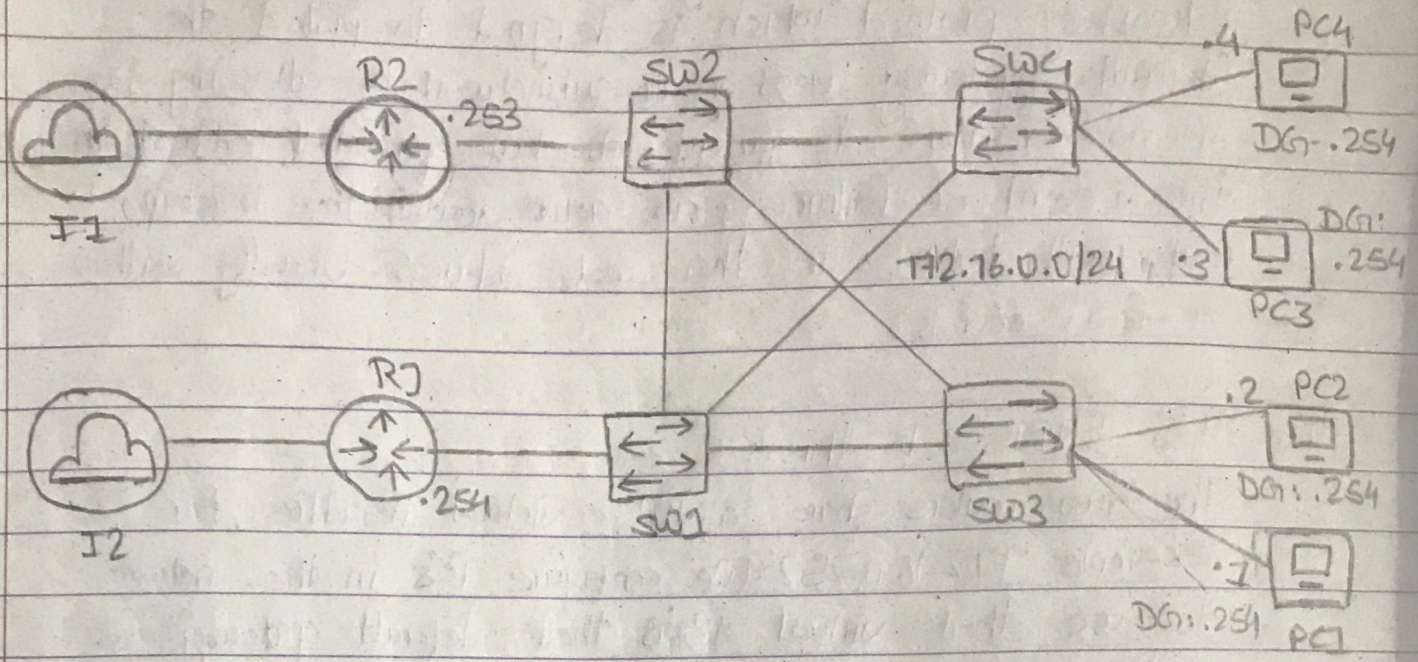
First Hop Redundancy Protocols.

#. Network diagram.



If connection o internet via I_2 fails PC2 can reach internet via alternate path SW3-SW2-I1. It could be slower & cheaper used only till when route to I_2 is recovered.

But its not that simple. Suppose, initially 172.16.0.254 is configured as default gateway. Each PC will have this add. as their default gateway. So anytime they need to send traffic outside local subnet, they send it to R1.

. What if R1 goes down due to some reason as hardware failure?

↳ There is backup router R2 connected to internet. But there'll still be problem. The PC's default gateway is still set to .254 (R1's address) Although R2 is available, PC don't know R1 is down & shall use R2. as default gateway.

Q How can we cause R2 to automatically become default gateway?

↳ That's the role of FHRP (First Hop Redundancy Protocol)

A first hop redundancy protocol (FHRP) is a computer networking protocol which is designed to protect the default gateway used on a subnetwork by allowing two or more routers to provide backup for that address; in the event of failure of an active router, the backup router will take over the act address. Usually within a few seconds.

How FHRPs do this?

↳ The two routers share a VIP, a virtual IP address, for example 172.16.0.252. We configure PCs in the network to use that virtual IP as their default gateway instead of actual IP of R1. Now, routers have to negotiate their roles with each other.

For this they send multicast "hello" mssg to each other. One will become active (say R1) while another (R2) becomes standby. The term 'Active' & 'Standby' may vary depending on which FHRP is being used.

Since, R1 is active, it'll act as default gateway for the network. If R1 fails, R2 will automatically become default gateway.

Now, if PC1 wants to send mssg outside of subnet (i.e internet) it knows default gateway IP but not MAC add. So, PC1 will make ARP request. Now since ARP is broadcast & both R1 & R2 have same IP (VIP). ~~What happ~~ which will send ARP reply?

Obviously R1. It sends unicast ARP reply.

Now PC1 knows both MAC & IP of default gateway.

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What happens if R1 goes down now?

↳ R2 won't receive hello mssg from R1. And after certain time it will become active router. All PCs have Virtual IP & MAC of default gateway, so we don't need to configure. Now R2 needs to make switches update their MAC address table.

To do this it sends "gratuitous ARP" replies.

Gratuitous ARP: ARP replies sent without being requested (no ARP req. mssg was received) The frames are broadcass to FFFF.FFFF.FFFF.

What if R1 comes online again? Will it become active again?

↳ No, it would be standby router instead. FHRPs are non-preemptiv The current active router won't actually give up its role, even if the former active role returns.

We can change this settings to make R1 (preempt) R2 and take back its active role automatically.

Summary

- Virtual IP is configured on two routers and a virtual MAC is generated for the virtual IP (each FHRP uses a different format for virtual MAC
- An active & standby router are elected.
- End hosts are configured to use Virtual IP as default gateway.

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HSRP (Hot Standby Router Protocol)

- Cisco proprietary.
- An active and standby router are elected.
- There are 2 versions: version 1 and version 2.
  Version 2 adds IPV6 support & increases the no. of groups that can be configured.

- Multicast IPV4 add: v1= 224.0.0.2
  v2= 224.0.0.102

- Virtual MAC add: v1= 0000.0c07.acXX (XX= HSRP group no)
  v2= 0000.0c9f.fXXX (XXX= HSRP group no.)

(Say in our earlier network (I), the group no. was 1 then)
Virtual MAC v1= 0000.0c07.ac01
v2= 0000.0c9f.f001.

- In a situation with multiple subnets/VLANs, you can configure a different active router in each each subnet/VLAN to load balance.
  Eg: Suppose our previous network diagram. Over there if PC 1 & PC3 are in one VLAN1 & subnet and PC2 & PC4 are in another subnet & VLAN2. Then we can set R1 as active for VLAN1 & standby for VLAN2. And, also, R2 will be standby for VLAN1 & active for VLAN2. Now traffic can be load balanced. :)

VRRP (Virtual Router Redundancy Protocol).

- Open standard.
- A master and backup router are elected.
- Multicast IPV4 address : 224.0.0.18
- Virtual MAC address : 0000.5e00.01xx (xx = VRRP group number.

{ Master → equivalent to Active of HSRP
{ Backup → equivalent to Standby of HSRP } 2.

If VRRP group number is 200, what'll be MAC address
Virtual MAC: 0000.5e00.01c8

- In a situation with multiple subnets/VLANS, we can configure a different master router in each subnet/VLAN to load balance same as that in HSRP.

{ recall VLAN ~~is~~ can't be 2 subnet. Subnet divide network at layer and VLAN at 2.

Gateway Load Balancing Protocol (GLBP).

- Cisco Proprietary.
- Load balances among multiple routers within a single subnet.
- An AVG (Active Virtual Gateway) is elected.
- Upto Four AVFs (Active Virtual forwarders) are assigned by the AVG (& the AVG itself can be AVF, too).

- Each AVF acts as default gateway for a portion of hosts in ~~network~~ subnet.

- Multicast IPv4 address : 224.0.0.102
- Virtual MAC address : 0007.b400.XXYY { XX = GLBP number, YY = AVF number }

#What is virtual MAC in AVF no 1 in GLBP group 1?

4) 0007.b400.0101

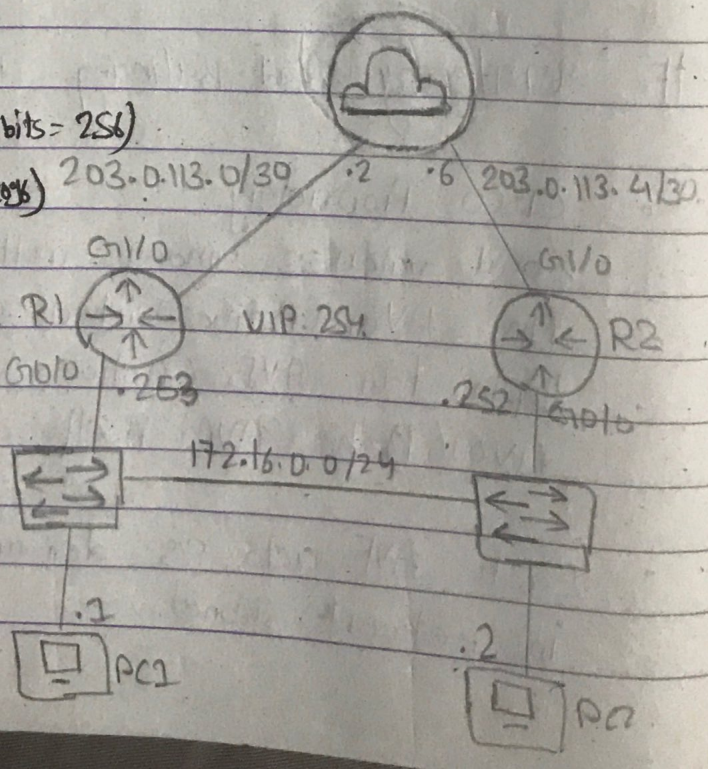
Comparing FHRPs.

FHRP	Terminology	Multicast IP	Virtual MAC	Cisco Proprietary
HSRP	Active / Standby	v1: 224.0.0.2 v2: 224.0.0.102	v1: 0000.0c07.acXX v2: 0000.0c9f.FXXX	Yes
VRRP	Master / Backup	224.0.0.18	0000.5e00.01XX	No.
GLBP	AVG/AVF	224.0.0.102	0007.b400.XXYY	Yes

Configuring HSRP (Configuring can't be asked on CCNA).

R1(config)# int g0/0
R1(config-if)# standby 1
<0-255> for HSRP v1 (2 bit hex = 8 bits = 256)
<0-4095> for HSRP v2 (3 hex = 12 bits = 4096)

Its not rule to match VLAN no.
(HSRP no. 9 is for convenience)
However group no. needs to match between routers.



R1(config-if)# standby 1 ip 172.16.0.254 (virtual ip)
R1(config-if)# standby 1 priority 200
R1(config-if)# standby 1 preempt. → only necessary on router we want to become active

What does priority do?
↳ Its used to determine active router.
The active router is determined in this order.
1→ Highest priority (default 100)
2→ highest IP address

Preempt- It causes router to take role of active router, even if another router the already has role

NOB: HSRP version 1 and version 2 aren't compatible. If R1 uses version 2 R2 also must use version 2.

Important commands.

show standby.